SOLUTIONS FOR WORKSHEET 6

Question 1. Prove that the following functions are differentiable on their domains.

(a) $f:\big(0,\infty\big)\to\mathbb{R}$ given by $f(x)=\sqrt{x+3}$.

(b) $g:\big(0,\infty\big)\to\mathbb{R}$ given by $g(x)=\frac{1}{x}$.

(c) $h:\big(0,\infty\big)\to\mathbb{R}$ given by $h(x)=4x+\frac{1}{x^2}$.

Proof. Let $a\in\big(0,\infty\big)$.

(a) $\lim\limits_{x\to a}\frac{f(x)-f(a)}{x-a}=\lim\limits_{x\to a}\frac{\sqrt{x+3}-\sqrt{a+3}}{x-a}\times\frac{\sqrt{x+3}+\sqrt{a+3}}{\sqrt{x+3}+\sqrt{a+3}}=\lim\limits_{x\to a}\frac{1}{\sqrt{x+3}+\sqrt{a+3}}=\frac{1}{2\sqrt{a+3}}$ (limit exists)

(b) $\lim\limits_{x\to a}\frac{g(x)-g(a)}{x-a}=\lim\limits_{x\to a}\frac{\frac{1}{x}-\frac{1}{a}}{x-a}=\lim\limits_{x\to a}\frac{a-x}{ax(x-a)}=\lim\limits_{x\to a}\frac{-1}{ax}=-\frac{1}{a^2}$ (limit exists)

(c) $\lim\limits_{x\to a}\frac{h(x)-h(a)}{x-a}=\lim\limits_{x\to a}\frac{4x+\frac{1}{x^2}-4a-\frac{1}{a^2}}{x-a}=4\times\lim\limits_{x\to a}\frac{x-a}{x-a}-\lim\limits_{x\to a}\frac{a+x}{a^2x^2}=4-\frac{2}{a^3}$ (limit exists)

Question 2. Suppose that $\cos x=\sum\limits_{n=0}^{\infty}a_nx^n$. By using Taylor theorem, find an explicit formula for the coefficients $a_n\in\mathbb{R}$.

Solution. Let $f(x)=\cos x$. Then $a_n=a_{2k}=\frac{(-1)^k}{(2k)!}\,\forall k\in\mathbb{N}\cup\{0\}$ (and $a_{\text{odd}}=0$) because

$$f^{(n)}(x)=\begin{cases}\cos x, & n\equiv 0\pmod 4\\ -\sin x, & n\equiv 1\pmod 4\\ -\cos x, & n\equiv 2\pmod 4\\ \sin x, & n\equiv 3\pmod 4\end{cases}\Rightarrow f^{(n)}(0)=\begin{cases}1, & n\equiv 0\pmod 4\\ 0, & n\equiv 1\pmod 4\\ -1, & n\equiv 2\pmod 4\\ 0, & n\equiv 3\pmod 4\end{cases}\Rightarrow f^{(2k)}(0)=(-1)^k.$$

Question 3. Using the ratio test, find R, the radius of convergence of the power series

(a) $\sum\limits_{k=0}^{\infty}(-1)^kx^k$ (b) $\sum\limits_{k=1}^{\infty}\frac{(-1)^{k+1}x^k}{k}$

Solution. (a) $\lim\left(\sqrt[k]{\left|(-1)^k\right|}\right)=1\Rightarrow R=1$.

(b) $\lim\left(\left|\frac{(-1)^{k+2}}{k+1}\div\frac{(-1)^{k+1}}{k}\right|\right)=\lim\left(\frac{k}{k+1}\right)=1\Rightarrow R=1$.

Question 4. Given that $\arctan x=\tan^{-1}x=\sum\limits_{k=0}^{\infty}\frac{(-1)^k}{2k+1}x^{2k+1}$ for $\big|x\big|\le 1$, find a power series expression for $\frac{1}{1+x^2}$ and its corresponding radius of convergence.

Solution. Knowing that $\frac{d}{dx}\tan^{-1}x=\frac{1}{1+x^2}$. Thus, $\frac{1}{1+x^2}=\sum\limits_{k=0}^{\infty}(-1)^kx^{2k}$ for $\big|x\big|<1$.

By Theorem 6.1, the differential has the same radius of convergence as the power series. Alternative: Use $\frac{1}{1+u}=\sum\limits_{k=0}^{\infty}(-1)^ku^k$ and substitution $u=x^2$ to obtain the result.

Question 5. Write down the Taylor polynomial of order 4, centred at $x_0 = 0$ for function

$$f : \left(-\tfrac{\pi}{2}, \tfrac{\pi}{2}\right) \to \mathbb{R} \text{ given by } f(x) = \sec x\,.$$

Solution. Let $p_4(x)$ be the Taylor polynomial of order 4, centred at $x_0 = 0$. Since

$$\begin{aligned}
f(x) &= \sec x && \Rightarrow f(0) = 1\\
f'(x) &= \sec x \tan x && \Rightarrow f'(0) = 0\\
f''(x) &= -\sec x + 2\sec^3 x && \Rightarrow f''(0) = 1\\
f'''(x) &= \tan x(-\sec x + 6\sec^3 x) && \Rightarrow f'''(0) = 0\\
f^{(4)}(x) &= \tfrac{-\sec x \tan x + 18\sec^3 x \tan x}{\cot x} + \tfrac{-\sec x + 6\sec^3 x}{\cos^2 x} && \Rightarrow f^{(4)}(0) = 5
\end{aligned}$$

It follows that $p_3(x) = 1 + \frac{1}{2}x^2 + \frac{5}{24}x^4$.

Question 6. Let $f_a : \mathbb{R} \to \mathbb{R}$ given by $f_a(x) = \begin{cases} x^a \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$, where $a \in \mathbb{R}$.

Find a value of a such that f_a is differentiable on $\mathbb{R}$ and

(a) f_a' is unbounded on $[0, 1]$.

(b) f_a' is continuous but not differentiable at 0.

(c) f_a' is differentiable on $\mathbb{R}$ but f_a'' is not continuous at 0.

Solution. $f_a'(x) = ax^{a-1}\sin\left(\frac{1}{x}\right) - x^{a-2}\cos\left(\frac{1}{x}\right),\ x \neq 0$.

$f_a''(x) = \left(a(a-1)x^2 - 1\right)x^{a-4}\sin\left(\frac{1}{x}\right) - 2(a-1)x^{a-3}\cos\left(\frac{1}{x}\right),\ x \neq 0$.

(a) Pick $a = 1.2$. (b) Pick $a = 2.2$. (c) Pick $a = 3.2$.

Question 7. Let $f : D \to \mathbb{R}$ be differentiable at the point $c \in D$. Prove that

$$\left(\lambda f\right)'(c) = \lambda f'(c) \text{ for each } \lambda \in \mathbb{R}\,.$$

Proof. Given $f'(c) = \lim\limits_{x\to c} \frac{f(x)-f(c)}{x-c}$. Let $g : D \to \mathbb{R}$ given by $g(x) = \lambda f(x)$. Then

$$g'(c) = \lim_{x\to c} \tfrac{g(x)-g(c)}{x-c} = \lim_{x\to c} \tfrac{\lambda f(x)-\lambda f(c)}{x-c} = \lambda \lim_{x\to c} \tfrac{f(x)-f(c)}{x-c} = \lambda f'(c)\,.$$

Question 8. Let $f, g : D \to \mathbb{R}$ be differentiable at $c \in D$ and $g(x) \neq 0\ \forall x \in D$. Prove that $\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{\left(g(c)\right)^2}$.

Proof. Use $\left(fg\right)'(c) = f'(c)g(c) + f(c)g'(c)$ and $\left(\frac{1}{g(c)}\right)' = -\frac{g'(c)}{(g(c))^2}$ to prove that

$$\left(\tfrac{f}{g}\right)'(c) = \left(f \times \tfrac{1}{g}\right)'(c) = f'(c) \times \tfrac{1}{g(c)} + f(c)\left(\tfrac{1}{g(c)}\right)' = \tfrac{f'(c)g(c) - f(c)g'(c)}{\left(g(c)\right)^2}\,.$$

Question 9. Prove the Interior Extremum Theorem.

Let $f : D \to \mathbb{R}$ be a function that is differentiable on $I = (a,b) \subseteq D$. Suppose f attains a maximum or minimum value at some point $c \in (a,b)$, then $f'(c) = 0$.

Proof. Case 1. Let f attain a maximum at $c \in (a,b)$. Then

$$f(x) \leq f(c) \quad \forall x \in (a,b).$$

Since f is differentiable on I so $f'(c)$ exists. Considering the right-hand one-sided (and left-hand one-sided) limit of the differential, then we obtain $f'(c) = 0$ because

$$f'(c) = \lim_{x \to c^+} \tfrac{f(x)-f(c)}{x-c} \leq 0, \text{ and}$$

$$f'(c) = \lim_{x \to c^-} \tfrac{f(x)-f(c)}{x-c} \geq 0.$$

The proof for when c is a minimum is very similar.

Case 2. Let f attain a minimum at $c \in (a,b)$. Then

$$f(x) \geq f(c) \quad \forall x \in (a,b).$$

Since f is differentiable on I so $f'(c)$ exists. Then $f'(c) = 0$ because

$$f'(c) = \lim_{x \to c^+} \tfrac{f(x)-f(c)}{x-c} \geq 0, \text{ and}$$

$$f'(c) = \lim_{x \to c^-} \tfrac{f(x)-f(c)}{x-c} \leq 0.$$

Question 10. (Theorem 6.1) Let $\sum_{n=0}^{\infty} c_n x^n$ be a power series with radius of convergence R. Prove that $\sum_{n=1}^{\infty} n c_n x^{n-1}$ also has radius of convergence R.

Proof. Suppose $\alpha = \lim \left| \frac{c_{n+1}}{c_n} \right| = \frac{1}{R}$, then $\lim \left| \frac{(n+1)c_{n+1}}{n c_n} \right| = \lim \left| \frac{n+1}{n} \right| \times \lim \left| \frac{c_{n+1}}{c_n} \right| = 1 \times \frac{1}{R} = \frac{1}{R}$. So, $\sum_{n=1}^{\infty} n c_n x^{n-1}$ also has radius of convergence R.

Question 11. Finish the proof of Theorem 7.4 (Lagrange's Remainder Theorem) from Chapter 5 of the notes.

Proof. Let $S_N(x) = \sum_{n=0}^{N} \frac{f^{(n)}(0)}{n!} x^n$, and $E_N(x) := f(x) - S_N(x)$. Note that

1. $E_N(x)$ is differentiable on (a,b) and continuous on $[a,b]$ as $f(x)$ and $S_N(x)$ do.
2. $E_N(0) = 0 \ \forall n \in \mathbb{N} \cup \{0\}$ because $S_N(x) = f(0) + \sum_{n=1}^{N} \frac{f^{(n)}(0)}{n!} x^n$.
3. Both functions x^{N+1} and $E_N(x)$ are differentiable on $(x,0)$ and continuous on $[x,0]$.

By CMVT, there exists $x_1 \in (x,0)$ such that

$$\frac{E_N'(x_1)}{(N+1)x_1^N} = \frac{E_N(0)-E_N(x)}{-x^{N+1}} = \frac{E_N(x)}{x^{N+1}}. \quad (1)$$

Follow the same pattern, $E_N'(x)$ and x^N are differentiable on $(x_1,0)$ and continuous on $[x_1,0]$. By CMVT, there exists $x_2 \in (x_1,0)$ such that

$$\frac{E_N''(x_2)}{Nx_2^{N-1}} = \frac{E_N'(0)-E_N'(x_1)}{-x_1^N} = \frac{E_N'(x_1)}{x_1^N}. \quad (2)$$

Carefully sub (1) into (2) to get

$$\frac{E_N''(x_2)}{N(N+1)x_2^{N-1}} = \frac{E_N'(x_1)}{(N+1)x_1^N} = \frac{E_N(x)}{x^{N+1}}. \quad (3)$$

Again, $E_N''(x)$ and x^{N-1} are differentiable on $(x_2,0)$ and continuous on $[x_2,0]$.
By CMVT, there exists $x_3 \in (x_2,0)$ such that

$$\frac{E_N'''(x_3)}{(N-1)x_3^{N-2}} = \frac{E_N''(x_2)}{x_2^{N-1}}. \quad (4)$$

Carefully sub (3) into (4) to get

$$\frac{E_N'''(x_3)}{(N+1)N(N-1)x_3^{N-2}} = \frac{E_N(x)}{x^{N+1}}. \quad (5)$$

Continuing this $N+1$ times result in the existence of an $x_{N+1} \in (x_N,0)$ such that

$$\frac{E_N(x)}{x^{N+1}} = \frac{E_N^{(N+1)}(x_{N+1})}{(N+1)!\,x_{N+1}^{N-N}} = \frac{E_N^{(N+1)}(x_{N+1})}{(N+1)!}.$$

Setting $c = x_{N+1} > x$ gives the

$$\frac{E_N(x)}{x^{N+1}} = \frac{E_N^{(N+1)}(c)}{(N+1)!} \Rightarrow E_N(x) = \frac{E_N^{(N+1)}(c)}{(N+1)!}x^{N+1}.$$

Now, $E_N(x) = f(x) - S_N(x)$ implies that $E_N^{(N+1)}(x) = f^{(N+1)}(x)$ since $S_N^{(N+1)}(x) = 0$ (because $S_N(x)$ is of degree N). Hence, $E_N(x) = \dfrac{f^{(N+1)}(c)}{(N+1)!}x^{N+1}$.

Question 12. Prove that $\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!}x^{2k}$ is true for all $x \in \mathbb{R}$.

Proof. $\left|E_N(x)\right| = \left|\frac{x^{N+1}}{(N+1)!}f^{(N+1)}(c)\right| \le \frac{|x|^{N+1}}{(N+1)!}$ since $\left|f^{(N+1)}(c)\right| \le 1$ for some c between 0 and x.
Now, $0 \le \left|E_N(x)\right| \le \frac{|x|^{N+1}}{(N+1)!}$ for all $x \in (-\pi,\pi)$. By Squeeze theorem, $\lim\left(\left|E_N(x)\right|\right) = 0$.
Since $\cos x = f(x) = f(x+2\pi)$ is a period function, it follows that $\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!}x^{2k}$ is true for all $x \in \mathbb{R}$. (Note: in Q2 we have found the Taylor series $\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!}x^{2k}$).

Question 13. Let $f:\mathbb{R}\to\mathbb{R}$ given by $f(x)=\begin{cases}\frac{1}{2}x & x\in\mathbb{Q}\\ x & x\notin\mathbb{Q}\end{cases}$. Is f differentiable at $x=0$?

Proof. $f(0+h)-f(0)=f(h)=\begin{cases}\frac{1}{2}h & x=h\in\mathbb{Q}\\ h & x=h\notin\mathbb{Q}\end{cases}\Rightarrow\frac{f(0+h)-f(0)}{h}=\frac{f(h)}{h}=\begin{cases}\frac{1}{2} & x=h\in\mathbb{Q}\\ 1 & x=h\notin\mathbb{Q}\end{cases}$.

Both rational and irrational numbers are arbitrarily close to 0, so $\lim_{h\to0}\frac{f(0+h)-f(0)}{h}$ does not exist. Hence, f is not differentiable at $x=0$.

Remark: If $\frac{f(a+h)-f(a)}{h}=\begin{cases}f_1(h) & x=h\in\mathbb{Q}\\ f_2(h) & x=h\notin\mathbb{Q}\end{cases}$, $\lim_{x\to a}f_1(h)=\lim_{x\to a}f_2(h)$. Then f is differentiable at $x=a$. For example, $f(x)=\begin{cases}x^2 & x\in\mathbb{Q}\\ 0 & x\notin\mathbb{Q}\end{cases}$ is differentiable at $x=0$.

Question 14. Let $f:D\to\mathbb{R}$ and f be differentiable on some interval $I\subset D$. Prove that if $f'(x)\neq0$ for each $x\in I$, then f is injective on I. Provide a counter example to demonstrate the converse is not true.

Proof. Let $a\neq b$ (without loss of its generality, let $a<b$). Suppose $f(a)=f(b)$ where $a,b\in I$. Now f is differentiable on (a,b) and continuous $[a,b]$, by Rolle Theorem, $f'(c)=0$ for some $c\in(a,b)$. But this contradicts to the fact that $f'(x)\neq0\ \ \forall x\in I$. Hence, $f(a)\neq f(b)$.

Alternative: Note that $f'(x)\neq0\ \forall x\in I\Rightarrow f'>0$ or $f'<0\ \forall x\in I$. This means that f is strictly increasing or decreasing on the interval I. Now suppose $a\neq b$, then without loss of its generality, let $a<b$. Since f is strictly increasing (decreasing), then we have $f(a)<f(b)\big(f(b)<f(a)\big)$, which means that $f(a)\neq f(b)$.

Counter example: Let $f(x)=x^3$ for $x\in\mathbb{R}$. Then f is injective but $f'(0)=0$ (existence of inflection point).

Question 15. Consider $\sum_{n=0}^{\infty}c_nx^n$ such that there exists $M>m>0$ with $m\le|c_n|<M$ $\forall n\in\mathbb{N}\cup\{0\}$. Prove that the radius of convergence for the power series is 1.

Proof. $m\le|c_n|<M\Rightarrow\sqrt[n]{m}\le\sqrt[n]{|c_n|}<\sqrt[n]{M}\ \ \forall n\in\mathbb{N}\cup\{0\}$. Since $\lim m^{\frac{1}{n}}=1=\lim M^{\frac{1}{n}}$, by the squeeze theorem, we have $\lim\sqrt[n]{|c_n|}=1$ and hence the radius of convergence for the power series is 1.

Question 16. Prove the L'Hôpital's Rule.

Assume f and g are continuous functions defined on an interval I containing a, and assume that f and g are differentiable on this interval, with the possible exception of the point a.

(a) Use the Cauchy Mean Value Theorem to provide a proof of the 0/0 case of L'Hôpital's Rule, i.e., if $f(a) = g(a) = 0$ and $g'(x) \neq 0$ for all $x \in I$, then

$$\lim_{x \to a} \frac{f'(x)}{g'(x)} = L \Rightarrow \lim_{x \to a} \frac{f(x)}{g(x)} = L\,.$$

(b) Prove that if $\lim_{x \to a} f(x) = \pm\infty$ and $\lim_{x \to a} g(x) = \pm\infty$, then

$$\lim_{x \to a} \frac{f'(x)}{g'(x)} = L \Rightarrow \lim_{x \to a} \frac{f(x)}{g(x)} = L\,.$$

Proof. (a) Case 0/0. Given $\lim_{x \to a} \frac{f'(x)}{g'(x)} = L$, i.e., $\lim_{x \to a^+} \frac{f'(x)}{g'(x)} = L = \lim_{x \to a^-} \frac{f'(x)}{g'(x)}$.

When $\lim_{x \to a^+} \frac{f'(x)}{g'(x)} = L$, i.e., for all $\varepsilon > 0$, there exists $\delta > 0$ such that $a < x < a + \delta \Rightarrow \left|\frac{f'(x)}{g'(x)} - L\right| < \varepsilon$ * $\forall x \in I$. Apply the CMVT on interval $\left(a, x\right)$ to obtain

$$\frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c)}{g'(c)} \text{ for some } c \in \left(a, x\right).$$

Since * is true for all $x \in I$, it follows that the inequality holds true at c. Hence,

$$\left|\frac{f(x)}{g(x)} - L\right| = \left|\frac{f(x) - f(a)}{g(x) - g(a)} - L\right| = \left|\frac{f'(c)}{g'(c)} - L\right| < \varepsilon\,.$$

This proves that $\lim_{x \to a^+} \frac{f(x)}{g(x)} = L$. Similarly, when $\lim_{x \to a^-} \frac{f'(x)}{g'(x)} = L$, we can prove that $\lim_{x \to a^-} \frac{f(x)}{g(x)} = L$, and thus implying $\lim_{x \to a} \frac{f(x)}{g(x)} = L$, as desired.

(b) Case ∞/∞. Write $f_1(x) = \frac{1}{f(x)}$ and $g_1(x) = \frac{1}{g(x)}$. Then the result follows.